

3.1 Selecting sourcetype

Field Extraction Report

Proof-of-Concept

1. Objective

This report documents a hands-on evaluation of Splunk Enterprise, covering the core workflow of a SIEM analyst: searching raw log data, structuring that data through field extraction, automating detection through alerts, and consolidating findings into dashboards. It also summarizes two applied log investigations carried out using the platform; the full methodology and evidence for those are documented.

2. Basic Search & Monitoring

Before building anything advanced, the first step in Splunk is simply asking it to show you what happened — similar to searching a mailbox by keyword, but here the “main” is every log line ingested into the platform.

```
index=main source="WinEventLog:Security"
```

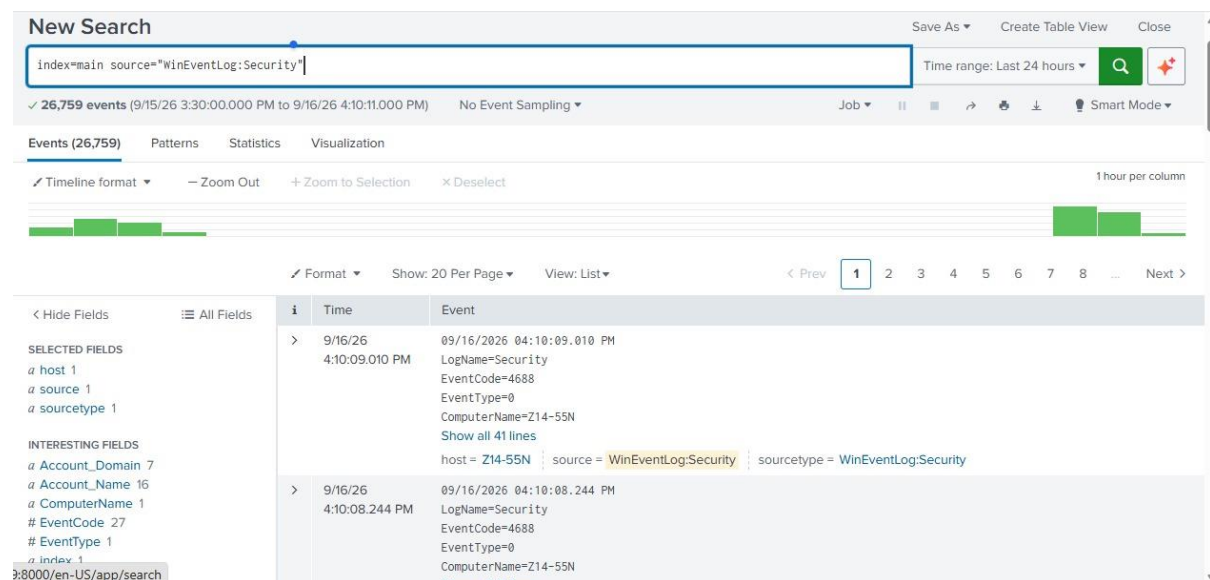


Figure 1 — 26,759 Windows Security events retrieved for the last 24 hours, with the built-in timeline showing event volume over time and interesting fields (Account_Domain, Account_Name, EventCode, etc.) automatically surfaced in the sidebar.

This confirms Windows Security event log data (process creation, logon events, etc.) is flowing into Splunk and is searchable, and demonstrates basic use of the search bar, time range picker, and field sidebar for initial triage.

3. Field Extraction

Raw logs often arrive as one long line of text. Field extraction teaches Splunk to recognize and pull out the meaningful parts of that text — like LogName or EventCode — as their own searchable columns, instead of requiring a full manual regular expression.

3.1 Selecting sourcetype

splunk> Enterprise

Extract Fields

Select Sample Select Method Select Fields Save Next >

Existing fields >

Select Sample Event

Choose a source or source type, select a sample event, and click Next to go to the next step. The field extractor will use the event to extract fields. [Learn more](#)

[I prefer to write the regular expression myself >](#)

Source type
WinEventLog:Application

Time Range
Last 90 days

Events

✓ 100 events (6/18/26 12:00:00.000 AM to 9/16/26 4:13:38.000 PM)

20 per page < Prev 1 2 3 4 5 Next >

filter Apply Sample: 1,000 events All events

```
09/16/2026 04:04:55.635 PM
LogName=Application
EventCode=16384
EventType=4
ComputerName=Z14-55N
SourceName=Microsoft-Windows-Security-SPP
Type=Information
RecordNumber=23989
Keywords=Classic
TaskCategory=None
OpCode=None
Message=Successfully scheduled Software Protection service for re-start at 2126-08-23T10:34:55Z. Reason: RulesEngine.
```

Figure 2 — Extract Fields wizard started against sourcetype `WinEventLog:Application`, with a 90-day sample window of 100 events to choose from.

3.2 Choosing the Sample Event

Extract Fields

Select Sample Select Method Select Fields Save Next >

Existing fields >

Select Sample Event

Choose a source or source type, select a sample event, and click Next to go to the next step. The field extractor will use the event to extract fields. [Learn more](#)

[I prefer to write the regular expression myself >](#)

Source type
WinEventLog:Application

Time Range
Last 90 days

```
09/16/2026 04:04:55.635 PM
LogName=Application
EventCode=16384
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RecordNumber=23989
Keywords=Classic
TaskCategory=None
OpCode=None
Message=Successfully scheduled Software Protection service for re-start at 2126-08-23T10:34:55Z. Reason: RulesEngine.
```

Figure 3 — A sample Application log event selected, with `LogName=Application` highlighted as the first field of interest.

3.1 Selecting sourcetype

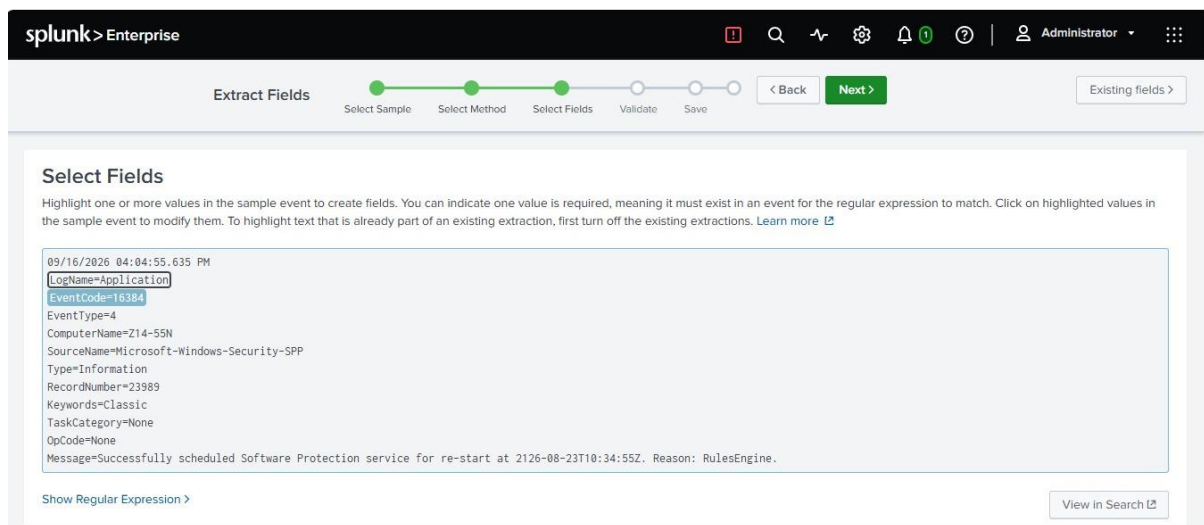


Figure 4 — EventCode=16384 additionally highlighted and selected, alongside LogName, to be extracted as a named field.

3.4 Save and Confirm

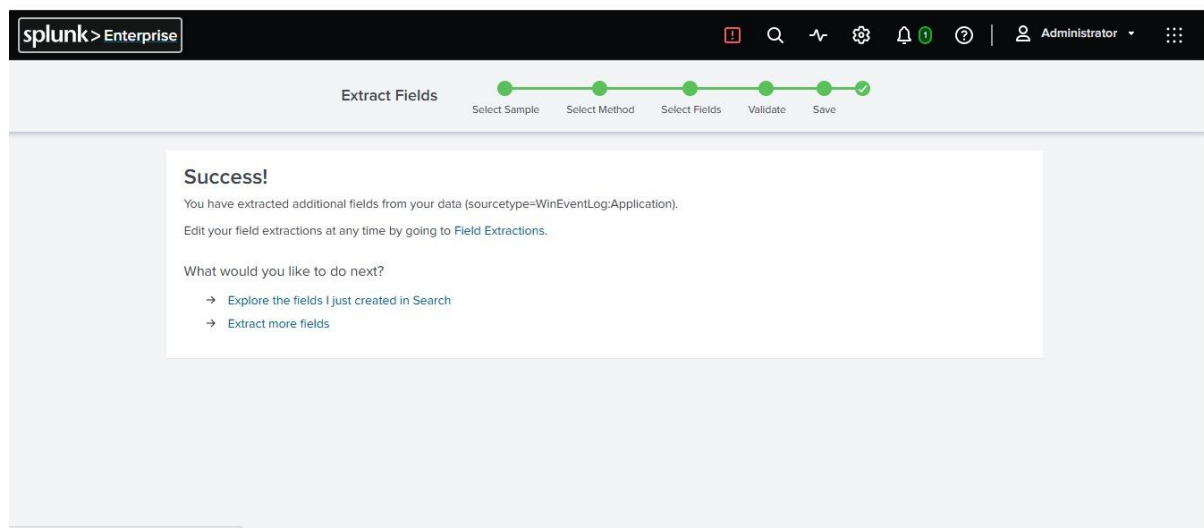


Figure 5 — Extraction saved successfully for sourcetype WinEventLog:Application, with the new fields immediately available for search.

This demonstrates the ability to turn unstructured log text into structured, filterable fields using Splunk's guided UI — the same underlying capability used with hand-written rex expressions elsewhere in this evaluation, but through the no-code workflow Splunk provides for analysts who prefer it.